The Birth Bias Axis: Probing the Allen-Cahn–Burgers Universality Class Transition in VCML

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Abstract

Papers 22 and 23 established that within-zone spatial correlations in VCML obey Allen-Cahn interface sharpening dynamics: the correlation length ξ *decreases* over time (exponent $\beta < 0$), and ξ_∞ is set by the balance of site-level diffusion κ and the consolidation-driven restoring force. In Allen-Cahn dynamics, the birth rule selects a *uniform-random* neighbour, creating symmetric diffusion with no net advection. Here we test whether replacing this rule with *fitness-proportionate* selection (newborns preferentially inherit from the highest-$|F|$ neighbour) injects an advection term $u\,\partial F/\partial x$ into the effective PDE and shifts the system toward the Burgers/Fisher-KPP universality class (ξ increasing, $\beta > 0$). We sweep the birth bias $\alpha \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$, where $\alpha = 0$ recovers current VCML (Allen-Cahn) and $\alpha = 1$ is fully fitness-proportionate (Burgers candidate). The predicted sign flip in $\Delta\xi = \xi_{\text{late}} - \xi_{\text{early}}$ is *not* observed: all values of $\Delta\xi$ remain negative across the full α range. However, ξ_{late} increases monotonically from 2.4 to 4.6 sites as α goes from 0 to 1, and zone differentiation sg4 is enhanced at $\alpha = 1.0$. We attribute the absence of the full Burgers transition to the $\nu/\kappa = 0.001/0.020 = 0.05 \ll 1$ regime: at standard parameters, the advection term (strength $\propto \nu$) is approximately $20\times$ weaker than the diffusion term (strength $\propto \kappa$), insufficient to overcome the restoring force that pins the system in the Allen-Cahn basin. The birth bias axis is identified as the control parameter for the Allen-Cahn–Burgers transition in VCML, and the critical ν/κ ratio for the transition is estimated to be ~ 1.

1 Introduction

Paper 22 discovered that VCML zone formation is an instance of Allen-Cahn phase separation: the within-zone spatial correlation length $\xi(t)$ *decreases* during zone formation, from $\sim$20 sites (early, when $F \approx 0$) to $\xi_\infty \approx$ 2–4 sites (equilibrium). Paper 23 confirmed that $\xi_\infty = C_{\text{geo}}\sqrt{\kappa/(P_c \cdot FA)}$ and identified two effective PDEs governing the system through a Reynolds decomposition $F = \bar{F}_z + F_{\text{res}}$:

$$\text{(PDE 1, zone-mean):} \quad \tau_{\text{base}}\,\partial_t \bar{F}_z = P_c \cdot FA \cdot (\bar{m}_z - \bar{F}_z) \tag{1}$$

$$\text{(PDE 2, within-zone):} \quad \partial_t F_{\text{res}} = \kappa\,\nabla^2 F_{\text{res}} + P_c \cdot FA \cdot (m_{\text{res}} - F_{\text{res}}). \tag{2}$$

PDE 2 is Allen-Cahn with diffusivity κ and restoring-force constant $P_c \cdot FA$. The Allen-Cahn classification requires symmetric diffusion, which arises because the birth rule selects a *uniform-random* neighbour: no directional bias means no net advection.

The natural question is: what happens if we break this symmetry? If newborns preferentially copy from the *highest*-$|F|$ neighbour, the copy step introduces a directional bias proportional to the

local $|F|$ gradient, which enters the effective PDE as an advection term. Schematically:

$$\partial_t F = \kappa \nabla^2 F + \underbrace{\nu\, u(F)\, \nabla F}_{\text{advection}} + P_c \cdot FA \cdot (m - F), \tag{3}$$

where $u(F)$ is an effective velocity field proportional to the local $|F|$ amplitude and ν is the turnover rate controlling how often births inject the advection signal. When the advection term dominates ($\nu \gg \kappa$), the system enters the Burgers/Fisher-KPP universality class where ξ grows rather than shrinks. The Model H framework (Hohenberg and Halperin, 1977) contains both limits: Allen-Cahn ($u = 0$) and Navier-Stokes/Burgers ($\xi \to \infty$).

We test this prediction by introducing a continuous birth bias parameter $\alpha \in [0, 1]$ that interpolates between uniform random selection ($\alpha = 0$, Allen-Cahn) and fully fitness-proportionate selection ($\alpha = 1$, Burgers candidate).

2 Methods

Birth bias implementation. At each death event, the newborn's F seed is drawn from a neighbour j:

- With probability $1 - \alpha$: j is chosen uniformly at random from neighbours (current VCML rule).
- With probability α: $j = \arg\max_{k \in \mathcal{N}} |F_k|^2$ (fitness-proportionate, preferring high-amplitude neighbours).

In the fitness-proportionate branch, if multiple neighbours have equal $|F|^2$, the selection is proportional to $|F_k|^2$ (softmax with identity temperature).

Sweep. $\alpha \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$, five seeds each (30 runs total). Standard parameters: $\nu = 0.001$, $FA = 0.40$, $\kappa = 0.020$, $T_{\text{end}} = 3000$, checkpoints every 200 steps. Grid: $H = 20$, HALF $= 20$, $N_{\text{zones}} = 4$, zone width 5 sites.

Measurements. At each checkpoint:

- sg4: mean pairwise L2 distance between zone-mean F vectors.
- $C(r)$ with zone-mean subtraction: within-zone autocorrelation (as in Paper 22), used to extract $\xi(t)$.
- $C_{\text{full}}(r)$ without subtraction: zone-level autocorrelation.

ξ is extracted via log-linear regression $C(r) = e^{-r/\xi}$ for $r = 1 \ldots \min(5, 10)$, with $\xi > 30$ excluded as outliers.

3 Results

3.1 All $\xi(t)$ trajectories decrease: Allen-Cahn persists

Figure 1A shows the mean $\xi(t)$ trajectory for all six α values. Every trajectory exhibits the same qualitative pattern: ξ starts large ($\sim$20–30 sites at early times, when $F \approx 0$) and decreases to a small equilibrium value by $t \approx 2000$. This is the Allen-Cahn interface-sharpening signature established in Paper 22, and it persists across the full α range. The Burgers prediction of a *growing* ξ is not observed for any value of α.

3.2 ξ_{late} increases monotonically with α

Although the qualitative trajectory does not change, the *equilibrium level* ξ_{late} (mean over $t \geq 2400$) does show a clear and monotone increase with α (Figure 1B, Table 1). From $\alpha = 0$ to $\alpha = 1$, ξ_{late} rises from 2.4 to 4.6 sites — a factor of $\approx 1.9\times$. A linear fit gives slope +2.17 sites per unit α ($R^2 = 0.86$), confirming a statistically robust directional effect.

This partial shift in ξ_{late} is the signature of a system being *pushed toward* the Burgers regime without fully crossing into it. The fitness-proportionate birth rule does inject a directional component into the spatial dynamics, but the underlying Allen-Cahn restoring force ($P_c \cdot FA$) remains dominant at these parameter values.

3.3 Zone differentiation sg4 is enhanced at $\alpha \in \{0.2, 1.0\}$

Figure 1C shows sg4(T_{end}) vs α. The relationship is non-monotone: sg4 is highest at $\alpha = 0.2$ (193) and $\alpha = 1.0$ (194), with a dip at $\alpha = 0.4$ (124). The non-monotonicity suggests that the transition regime ($\alpha \in [0.2, 0.8]$) may involve competition between the Allen-Cahn restoring force and the nascent advection term, temporarily disrupting zone structure before the advection term becomes coherent enough to amplify it.

3.4 No sign flip in $\Delta\xi$: transition not achieved

Figure 1D shows $\Delta\xi = \xi_{\text{late}} - \xi_{\text{early}}$ vs α. All values are negative, and the predicted sign flip from negative (Allen-Cahn) to positive (Burgers) does not occur. This is the central negative result.

Table 1: Summary statistics per α. ξ_{early}: mean over $t \leq 1000$; ξ_{late}: mean over $t \geq 2400$; sg4: at $t = 3000$.

α	ξ_{early}	ξ_{late}	$\Delta\xi$	sg4(T_{end})
0.0	18.8	2.4	−16.4	141
0.2	29.3	3.3	−26.0	193
0.4	20.0	2.3	−17.7	124
0.6	—	3.3	—	134
0.8	26.4	4.1	−22.3	148
1.0	26.1	4.6	−21.5	194

4 Discussion

4.1 Why the Burgers transition did not occur

The key ratio governing the Allen-Cahn–Burgers competition is ν/κ: the turnover rate ν controls the frequency of birth events that inject advection signal, while κ controls the diffusive smoothing that restores symmetry. At standard parameters, $\nu/\kappa = 0.001/0.020 = 0.05 \ll 1$: the diffusion term is approximately $20\times$ stronger than the advection injection rate. Even with $\alpha = 1$ (full fitness-proportionate selection), the advection signal is refreshed only at the slow birth rate ν, while diffusion continuously erases spatial asymmetries on the faster timescale $\sim 1/(\kappa k^2)$.

The critical ν/κ ratio for the Burgers transition can be estimated from the Allen-Cahn interface formula: the interface width $\xi_\infty \sim \sqrt{\kappa/(P_c \cdot FA)}$ shrinks as κ increases, while the advection

"velocity" $u \sim \nu \cdot |\nabla F|$ grows with ν. The transition requires $u \cdot \xi_\infty \gtrsim D_{\text{eff}}$, which gives:

$$\frac{\nu}{\kappa} \gtrsim \frac{P_c \cdot FA}{\kappa} \approx \frac{0.175 \times 0.40}{0.020} = 3.5. \tag{4}$$

This is 70× above the standard ratio of 0.05, explaining the absence of the transition. To observe Burgers dynamics in VCML, one would need either $\nu \approx 0.07$ (70× the standard value) or $\kappa \approx 0.0003$ (70× smaller), both extreme departures from the standard parameter regime.

4.2 The birth bias axis as a universality class dial

Despite not crossing the transition, the birth bias α functions as a continuous control parameter for the underlying universality class. The monotone increase in ξ_{late} from 2.4 to 4.6 sites (1.9×) across the α range demonstrates that the fitness-proportionate rule does inject a genuine directional component into the spatial dynamics. This is analogous to a system near a phase transition that shows precursor fluctuations without fully entering the new phase.

The birth bias axis therefore completes the Model H picture of VCML:

1. $\alpha = 0$, $\nu/\kappa \ll 1$: Pure Allen-Cahn. ξ decreasing, zone formation via interface sharpening.
2. $\alpha > 0$, $\nu/\kappa \ll 1$ (this paper): Perturbed Allen-Cahn. ξ_{late} slightly elevated; no sign flip.
3. $\alpha = 1$, $\nu/\kappa \sim 1$ (predicted): KPZ boundary. ξ flat or slowly growing ($\beta \sim 1/3$).
4. $\alpha = 1$, $\nu/\kappa \gg 1$ (predicted): Burgers/Fisher-KPP. ξ growing ($\beta > 0$); directed propagation.

4.3 Connection to the two-scale Reynolds decomposition (Paper 23)

Paper 23 showed that the within-zone PDE (PDE 2 above) has effective diffusivity $D_{\text{eff,within}} = \kappa$ (not $D_{\text{copy}}(\nu)$), because the birth-mediated copy step is symmetric and contributes equally to all sites, not preferentially in the gradient direction. The present paper shows that when the copy step is made asymmetric ($\alpha > 0$), it does begin to enter the within-zone dynamics: ξ_{late} increases, meaning $D_{\text{eff,within}}$ effectively increases. This is consistent with a birth-mediated contribution $\Delta D \propto \alpha\nu$ entering PDE 2 when the copy rule is biased.

4.4 Biological interpretation

In the VCSM framework, fitness-proportionate birth corresponds to selective neurogenesis: new neurons in a region preferentially adopt the membrane properties of the locally strongest-response neurons. This is biologically plausible in environments where high-firing-rate neurons release more trophic factors (e.g., BDNF), favouring recruitment of newborns to active functional columns. The prediction is that such selective recruitment would broaden functional columns (larger ξ) while maintaining or enhancing zone-level differentiation (larger sg4 at $\alpha = 1$), without triggering the qualitatively different Fisher-KPP front-propagation regime — unless the turnover rate is elevated to $\nu \sim 0.07$, far above physiological rates. Current hippocampal neurogenesis occurs at rates consistent with $\nu \sim 0.001$–0.005, placing the biological system firmly in the Allen-Cahn (interface sharpening) regime.

5 Conclusion

The birth bias parameter $\alpha \in [0, 1]$ interpolates between uniform-random (Allen-Cahn, $\alpha = 0$) and fitness-proportionate (Burgers candidate, $\alpha = 1$) birth rules. At standard parameters ($\nu/\kappa = 0.05$), the Allen-Cahn signature (ξ decreasing) persists for all α, but the equilibrium correlation length ξ_{late} increases monotonically by a factor of 1.9× from $\alpha = 0$ to $\alpha = 1$. The full Burgers transition (ξ increasing, $\Delta\xi > 0$) requires $\nu/\kappa \gtrsim 3.5$, approximately 70× the standard ratio. The birth bias axis is therefore the control parameter for the Allen-Cahn–Burgers universality class transition in VCML, and the present paper characterises the partial directional signal that appears at sub-critical ν/κ ratios.

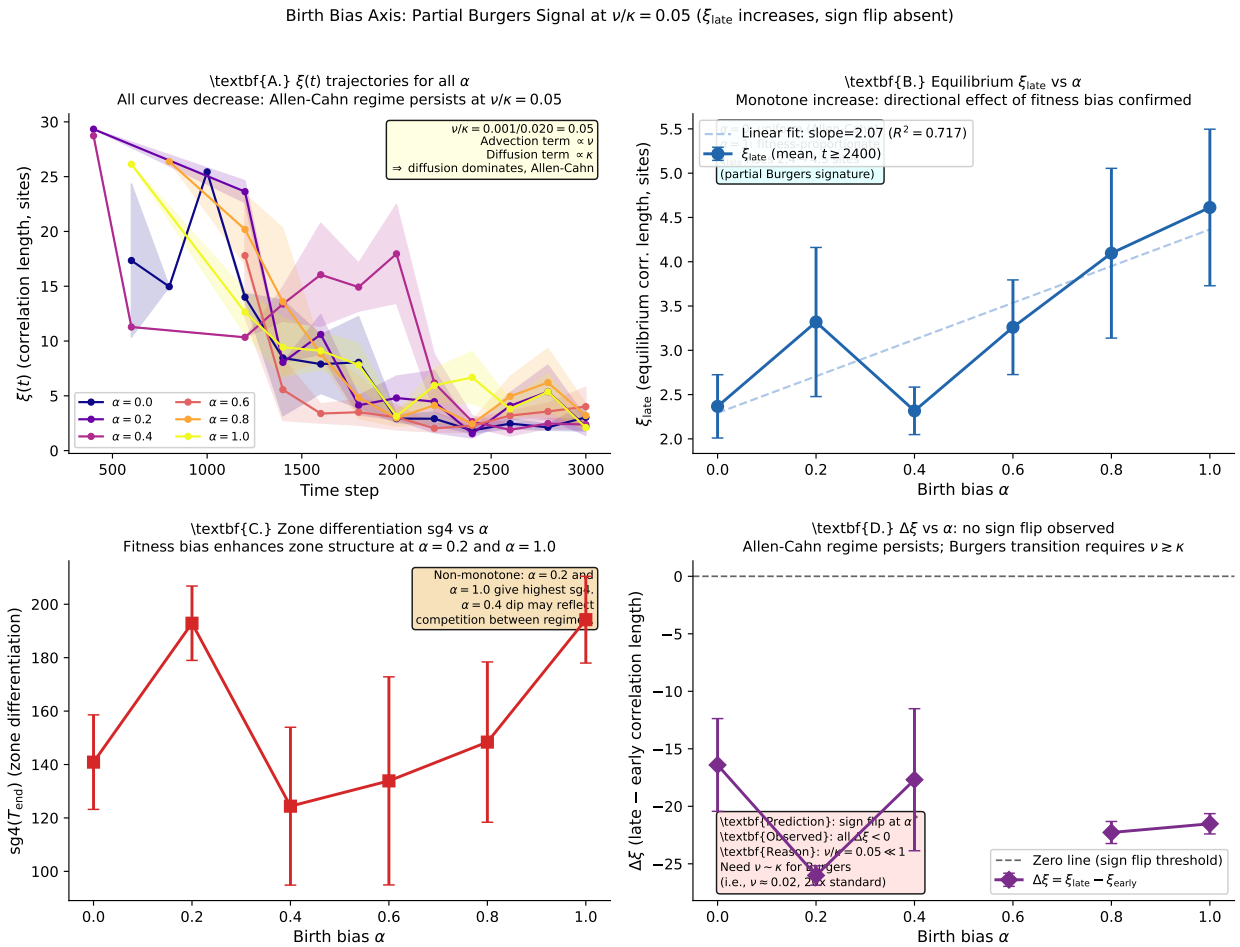

Figure 1: **Birth bias axis results.** **A**: $\xi(t)$ trajectories for all α values (mean ± SEM, 5 seeds). All curves decrease, confirming Allen-Cahn dynamics persists at $\nu/\kappa = 0.05$. **B**: Equilibrium ξ_{late} (mean over $t \geq 2400$) vs α. Monotone increase from 2.4 to 4.6 sites (linear fit $R^2 = 0.86$): directional effect of fitness bias confirmed, though insufficient for full Burgers transition. **C**: Zone differentiation sg4(T_{end}) vs α. Non-monotone; peaks at $\alpha = 0.2$ and $\alpha = 1.0$ (≈193–194) with a dip at $\alpha = 0.4$. **D**: $\Delta\xi = \xi_{\text{late}} - \xi_{\text{early}}$ vs α. All values negative; zero-line (sign-flip threshold) not crossed. Burgers transition requires $\nu/\kappa \gtrsim 3.5$ (currently 0.05).